NovaTech Solutions Pvt. Ltd.

NovaDesk Infrastructure Guide

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Document Control

Field	Value
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1. Introduction

This guide describes the infrastructure architecture supporting the NovaDesk Enterprise Service Management Platform.

It provides operational guidance for:

- Platform Engineers
- DevOps Engineers
- Site Reliability Engineers (SRE)
- Database Administrators
- Infrastructure Architects
- Technical Leads

Topics covered include:

- Cloud infrastructure
- Networking
- Compute resources
- Storage
- High availability
- Kubernetes
- Monitoring
- Security
- Disaster recovery

Application development standards are documented separately in the **NovaDesk Developer Guide (TECH-DEV-001).**

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2. Infrastructure Overview

NovaDesk is deployed using a cloud-native architecture designed for scalability, resilience, and operational efficiency.

The infrastructure is built around containerized workloads orchestrated by Kubernetes.

Primary design goals include:

- High availability
- Horizontal scalability
- Fault tolerance
- Automated deployment

- Secure networking
- Infrastructure as Code
- Continuous monitoring

Core Infrastructure Components

The production platform consists of:

- Kubernetes Cluster
- Docker Containers
- PostgreSQL Database
- Redis Cache
- Elasticsearch Cluster
- Object Storage
- Load Balancer
- Reverse Proxy
- Monitoring Platform
- Logging Platform

Each component is independently scalable.

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3. Architecture Overview

NovaDesk follows a layered cloud architecture.

```
      Internet
         |
         ▼
   DNS Resolution
         |
         ▼
External Load Balancer
         |
         ▼
 Reverse Proxy Layer
```

```
         │
         ▼
   Kubernetes Ingress
         │
   ┌─────────────┼─────────────┐
   ▼     ▼     ▼
Frontend Pods   API Pods   Background Workers
   │     │     │
   └─────────────┼─────────────┘
         ▼
   Internal Services
         │
   ┌───────────┼───────────┐
   ▼     ▼     ▼
PostgreSQL   Redis   Elasticsearch
```

All external traffic passes through the load balancer before reaching application services.

Infrastructure Principles

The platform follows these operational principles:

- Stateless application services
- Immutable container images
- Automated deployments
- Infrastructure as Code
- Health-based routing
- Centralized logging
- Centralized monitoring

These principles simplify scaling and recovery.

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4. Environment Layout

NovaDesk maintains multiple isolated environments.

Environment	Purpose
Development	Feature implementation
Integration	Shared testing
Quality Assurance	Functional validation
Staging	Production simulation
Production	Customer workloads

Each environment has separate compute, storage, databases, and configuration.

No production customer data is copied into lower environments without formal approval and sanitization.

Production Environment

Production infrastructure provides:

- High Availability
- Automated backups
- Continuous monitoring
- Multi-zone deployment
- Disaster recovery capability
- Auto scaling

Only approved release pipelines may deploy to production.

Staging Environment

Staging closely mirrors production.

It is used for:

- Release validation
- Performance testing
- User acceptance testing
- Deployment rehearsal

Configuration differences between staging and production are minimized.

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5. Network Topology

The production environment is segmented into multiple network zones.

Internet

↓

Public Subnet

↓

Load Balancer

↓

Private Application Subnet

↓

Database Subnet

Database servers are never directly accessible from the public internet.

Network Segmentation

Network zones include:

- Public Network
- Application Network
- Database Network
- Management Network

Traffic between network zones is restricted by firewall policies.

Firewall Rules

Default policy:

- Deny all inbound traffic.
- Allow only required application ports.
- Restrict database access to application services.
- Restrict administrative access.

Firewall rules are reviewed quarterly.

Internal Communication

Internal services communicate over encrypted private networking.

Service-to-service traffic is authenticated and encrypted where supported.

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6. Compute Infrastructure

NovaDesk workloads execute inside Kubernetes-managed containers.

Worker Nodes

Worker nodes host application workloads.

Typical responsibilities include:

- Frontend services
- REST API services
- Background processing
- Scheduled jobs

Control Plane

The Kubernetes control plane manages:

- Scheduling
- Service discovery
- Cluster health
- API server
- Configuration management

Administrative access to the control plane is restricted.

Resource Allocation

Each workload specifies:

- CPU requests
- CPU limits
- Memory requests
- Memory limits

Resource quotas prevent a single application from consuming excessive cluster capacity.

Auto Scaling

Horizontal Pod Autoscaling (HPA) is enabled for supported services.

Scaling decisions consider:

- CPU utilization
- Memory utilization
- Request throughput
- Queue depth

Autoscaling policies are reviewed after significant workload changes.

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7. Storage Architecture

NovaDesk uses different storage technologies depending on workload requirements.

Persistent Storage

Persistent storage is used for:

- PostgreSQL
- Elasticsearch
- Uploaded documents
- Application configuration

Persistent volumes remain available after container restarts.

Object Storage

Object storage contains:

- Attachments
- Knowledge Base documents
- Exported reports
- Backup archives

Objects are versioned and protected through lifecycle policies.

Cache Storage

Redis provides:

- Session caching
- Frequently accessed data
- Temporary application state

Cached data is considered disposable and is not included in long-term backup strategies.

Storage Classes

Storage Type	Typical Use
SSD	Database workloads
Standard Persistent Disk	Application storage
Object Storage	Documents and backups
Ephemeral Storage	Temporary container data

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8. DNS and Load Balancing

External users access NovaDesk using corporate DNS records.

DNS

DNS provides:

- Name resolution
- High availability
- Failover support

Example endpoint:

https://support.novatechsolutions.com

DNS changes follow the organization's approved change management process.

Load Balancer

The external load balancer distributes traffic across healthy application instances.

Capabilities include:

- SSL termination
- Health checks
- Session affinity (where required)
- Traffic distribution
- Automatic failover

Only healthy application instances receive production traffic.

Health Checks

Health checks execute every **30 seconds**.

Failed instances are automatically removed from the load balancer rotation.

Instances are returned to service only after passing consecutive health checks.

Traffic Distribution

Requests are distributed using load-balancing algorithms designed to maintain consistent application performance.

Long-running background jobs are isolated from interactive user traffic.

Revision History

Version	Date	Description
1.0	January 2024	Initial Infrastructure Guide
2.0	August 2025	Added Kubernetes Architecture
3.0	January 2026	Updated Environment Layout, Networking, and Storage Standards

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9. Kubernetes Cluster

NovaDesk is deployed on a Kubernetes cluster that orchestrates application containers across multiple worker nodes.

The cluster provides:

- Automated scheduling
- Self-healing
- Service discovery
- Horizontal scaling
- Rolling deployments
- Resource isolation

Cluster Components

Component	**Purpose**
API Server	Cluster management interface
Scheduler	Assigns workloads to nodes
Controller Manager	Maintains desired cluster state
etcd	Stores cluster configuration
Worker Nodes	Execute application workloads
Kubelet	Manages containers on each node
kube-proxy	Handles service networking

Namespaces

NovaDesk separates workloads using Kubernetes namespaces.

Namespace	Purpose
frontend	Web application
backend	REST API services
workers	Background processing
monitoring	Monitoring components
logging	Log aggregation

infrastructure Shared platform services

Namespaces improve security and operational isolation.

Deployments

Application workloads are managed through Kubernetes Deployments.

Deployment features include:

- Replica management
- Rolling updates
- Automatic rollback
- Version tracking
- Health monitoring

Replica counts are adjusted based on workload requirements.

Services

Internal communication is handled through Kubernetes Services.

Service types used:

- ClusterIP
- LoadBalancer
- ExternalName (limited use)

NodePort services are not used in production.

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10. Docker Infrastructure

NovaDesk packages every application component as a Docker container.

Container Images

Separate images exist for:

- Frontend
- Backend API
- Background Workers
- Scheduled Jobs

- Reporting Services

Each image is independently versioned.

Image Standards

Container images must:

- Use approved base images.
- Run as non-root users.
- Expose only required ports.
- Avoid unnecessary packages.
- Pass vulnerability scanning.

Images are rebuilt whenever security patches become available.

Container Registry

Images are stored in the organization's private container registry.

Only signed and approved images may be deployed to production.

Old image versions are retained according to the container retention policy.

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11. Ingress and Reverse Proxy

External requests enter the platform through a reverse proxy and Kubernetes Ingress.

Responsibilities include:

- Request routing
- TLS termination
- HTTP to HTTPS redirection
- Security headers
- Rate limiting
- Compression

URL Routing

Typical routing configuration:

/ → Frontend

/api → Backend Services

/docs → Documentation Portal

/health → Health Endpoint

Routing rules are managed through version-controlled infrastructure configuration.

HTTP Headers

The reverse proxy automatically adds security-related headers, including:

- Strict-Transport-Security
- X-Content-Type-Options
- X-Frame-Options
- Referrer-Policy
- Content-Security-Policy

These headers improve browser-side security.

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12. SSL/TLS Configuration

All external communication uses encrypted HTTPS connections.

TLS Requirements

NovaDesk supports:

- TLS 1.2
- TLS 1.3

Older protocol versions are disabled.

Certificates

Certificates are:

- Digitally signed
- Automatically renewed

- Continuously monitored

Certificate expiration alerts are generated **30 days** before expiration.

Secure Communication

Encryption is required for:

- Client to Load Balancer
- Load Balancer to Ingress
- Internal service communication where supported
- Database connections
- Administrative access

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13. Secrets Management

Sensitive configuration values are stored separately from application code.

Examples of Secrets

Secrets include:

- Database passwords
- API keys
- JWT signing keys
- Encryption keys
- OAuth client credentials
- SMTP credentials

Secrets must never be committed to source control.

Secret Rotation

Production secrets are rotated:

Secret Type	Rotation Interval
Database Passwords	Every 180 Days
API Keys	Every 90 Days

TLS Certificates	Before Expiration
JWT Signing Keys	Annually

Emergency rotation may occur following a security incident.

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14. Monitoring

Continuous monitoring provides visibility into application and infrastructure health.

Monitoring Platform

The monitoring stack collects metrics from:

- Kubernetes
- PostgreSQL
- Redis
- Elasticsearch
- Application Services
- Load Balancer
- Operating System

Dashboards are maintained using Grafana.

Key Metrics

The following metrics are monitored:

Metric	Description
CPU Utilization	Processor usage
Memory Utilization	Memory consumption
Disk Utilization	Storage usage
Request Throughput	Requests per second
Response Time	API latency
Error Rate	Failed requests
Pod Restarts	Container stability

Database Connections Active connections

Metrics are retained for **90 days** for operational analysis.

Alert Thresholds

Alerts are generated when:

- CPU utilization exceeds **85%**.
- Memory utilization exceeds **90%**.
- Disk usage exceeds **80%**.
- API error rate exceeds **5%**.
- Pod restart frequency exceeds configured thresholds.

Alert thresholds are reviewed quarterly.

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15. Centralized Logging

All application and infrastructure logs are collected centrally.

Log Sources

Logs are collected from:

- Application containers
- Kubernetes nodes
- PostgreSQL
- Redis
- Elasticsearch
- Load Balancer
- Reverse Proxy
- Operating System

Log Structure

Every log entry includes:

- Timestamp
- Service Name

- Environment
- Log Level
- Correlation ID
- Host Name
- Message

Structured logging simplifies searching and troubleshooting.

Log Retention

Environment	Retention
Development	30 Days
Staging	90 Days
Production	365 Days

Archived logs remain available according to organizational retention policies.

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16. High Availability

NovaDesk is designed to minimize service interruptions.

High Availability Strategy

Availability is achieved through:

- Multiple application replicas
- Multi-zone deployment
- Automatic failover
- Load balancing
- Database replication
- Health-based routing

Pod Health Checks

Every application pod exposes:

- Liveness Probe
- Readiness Probe

Failed liveness checks restart unhealthy containers automatically.

Failed readiness checks remove instances from traffic until recovery.

Database Availability

Production databases provide:

- Primary instance
- Standby replica
- Automated failover
- Continuous replication

Replication health is monitored continuously.

Maintenance Windows

Planned infrastructure maintenance is scheduled during approved maintenance windows.

Standard maintenance notification:

Minimum 7 days in advance

Emergency maintenance may occur immediately if required to protect security or platform stability.

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17. Infrastructure Best Practices

Platform Engineering teams should:

- Use Infrastructure as Code for all infrastructure changes.
- Apply the principle of least privilege.
- Encrypt data in transit and at rest.
- Monitor infrastructure continuously.
- Test failover procedures regularly.
- Keep container images updated.
- Rotate secrets according to policy.
- Remove unused infrastructure resources.
- Review capacity metrics monthly.
- Document all production changes.

Following these practices improves reliability, scalability, and security.

18. Disaster Recovery

NovaDesk maintains a documented Disaster Recovery (DR) strategy to ensure business continuity during major infrastructure failures.

The objectives of the DR program are to:

- Restore critical services rapidly.
- Minimize data loss.
- Protect customer information.
- Maintain service availability.
- Meet Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets.

Disaster Recovery Objectives

Environment	RTO	RPO
Production	2 Hours	15 Minutes
Staging	4 Hours	4 Hours
Development	8 Hours	24 Hours

Disaster Scenarios

Recovery procedures exist for:

- Complete data center outage
- Cloud region failure
- Kubernetes cluster failure
- Database corruption
- Storage system failure
- Ransomware attack
- Network outage
- DNS service failure

Each scenario has a documented recovery runbook maintained by the Platform Engineering Team.

Recovery Workflow

Standard recovery process:

1. Detect incident.
2. Declare disaster.
3. Notify stakeholders.
4. Activate Disaster Recovery Team.
5. Restore infrastructure.
6. Restore databases.
7. Validate application functionality.
8. Resume customer access.
9. Conduct post-incident review.

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19. Scaling Strategy

NovaDesk is designed to scale horizontally to accommodate increasing workloads.

Horizontal Scaling

Application services scale by increasing the number of running pods.

Scaling is triggered by:

- CPU utilization
- Memory utilization
- Request throughput
- Queue depth

Horizontal scaling minimizes downtime during traffic spikes.

Vertical Scaling

When necessary, workloads may also receive additional resources.

Examples include:

- Additional CPU
- Additional memory
- Larger database instances

- Increased storage capacity

Vertical scaling is planned during approved maintenance windows.

Database Scaling

PostgreSQL scaling strategies include:

- Read replicas
- Connection pooling
- Query optimization
- Index tuning
- Partitioning for large tables

Database scaling decisions are based on observed workload characteristics.

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20. Capacity Planning

Capacity planning ensures infrastructure can support expected business growth.

Capacity Reviews

Platform Engineering conducts capacity reviews:

Resource	Review Frequency
CPU	Monthly
Memory	Monthly
Storage	Weekly
Database Growth	Monthly
Network Utilization	Monthly

Growth Forecasting

Capacity planning considers:

- Active user growth
- Ticket volume
- Knowledge Base growth
- Asset inventory expansion

- Attachment storage
- API request volume

Forecasts are reviewed quarterly and adjusted as business needs evolve.

Scaling Thresholds

Infrastructure expansion is evaluated when:

- Average CPU utilization exceeds **70%** for 30 consecutive days.
- Storage utilization exceeds **75%**.
- Average API response time exceeds established service objectives.
- Database growth trends indicate capacity limits within the next six months.

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21. Infrastructure Maintenance

Routine maintenance helps ensure infrastructure stability and security.

Daily Activities

Platform Engineers should:

- Review monitoring dashboards.
- Confirm backup completion.
- Review infrastructure alerts.
- Verify cluster health.
- Check replication status.

Weekly Activities

- Review storage capacity.
- Validate backup retention.
- Inspect failed deployments.
- Review system logs.
- Confirm certificate validity.

Monthly Activities

- Apply operating system patches.
- Update container base images.
- Review security advisories.
- Validate monitoring configuration.
- Review infrastructure costs.

Quarterly Activities

- Conduct disaster recovery exercises.
- Test failover procedures.
- Review firewall rules.
- Validate access permissions.
- Audit infrastructure documentation.

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22. Security Hardening

Infrastructure components should follow secure configuration baselines.

Operating System Security

Recommended practices:

- Apply security patches promptly.
- Disable unused services.
- Remove unnecessary packages.
- Enforce strong authentication.
- Restrict administrative access.

Kubernetes Security

Cluster security includes:

- Role-Based Access Control (RBAC)
- Network Policies
- Pod Security Standards
- Image vulnerability scanning

- Admission controller validation

Only approved container images may be deployed.

Network Security

Network protections include:

- Firewall rules
- Private subnets
- TLS encryption
- Secure administrative access
- DDoS protection
- Traffic monitoring

Network configurations are reviewed after significant infrastructure changes.

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23. Troubleshooting Guide

The following table summarizes common infrastructure issues.

Issue	Possible Cause	Recommended Resolution
Pod CrashLoopBackOff	Application startup failure	Review application logs and configuration
High API latency	Increased workload	Review autoscaling metrics and database performance
Database connection failures	Database unavailable	Verify database health and connectivity
SSL certificate warning	Certificate nearing expiration	Renew certificate before expiry
Failed deployment	Configuration or image issue	Roll back to the previous stable release
High storage utilization	Rapid data growth	Expand storage and archive obsolete data

Persistent issues should be escalated to the Platform Engineering Team.

Incident Response

Priority 1 infrastructure incidents include:

- Complete production outage
- Kubernetes control plane failure
- Database unavailability
- Load balancer failure
- Major security incident

Response actions:

1. Notify on-call engineer.
2. Engage Incident Manager.
3. Assemble response team.
4. Restore service.
5. Perform root cause analysis.
6. Document corrective actions.

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24. Operational Checklist

Before every production deployment:

- Infrastructure health verified.
- Monitoring dashboards operational.
- Backups completed successfully.
- Rollback plan documented.
- Release notes approved.
- Database migrations reviewed.
- Stakeholders notified.
- Maintenance window confirmed.

Post-Deployment Validation

After deployment:

- Verify application availability.
- Confirm health endpoints.
- Review monitoring dashboards.
- Validate log collection.

- Execute smoke tests.
- Monitor error rates.
- Confirm customer access.

Deployments remain under enhanced monitoring for the first **60 minutes** after completion.

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25. Operational Best Practices

Platform Engineering teams should:

- Automate infrastructure provisioning.
- Use version-controlled Infrastructure as Code.
- Monitor systems continuously.
- Test disaster recovery regularly.
- Review capacity trends proactively.
- Rotate secrets according to policy.
- Keep operating systems patched.
- Remove obsolete resources.
- Review infrastructure costs quarterly.
- Document operational procedures.

Continuous operational improvement reduces risk and improves service reliability.

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26. Frequently Asked Questions

Q1. Which environments are maintained for NovaDesk?

Development, Integration, Quality Assurance, Staging, and Production.

Q2. What are the production recovery objectives?

Production targets are an **RTO of 2 hours** and an **RPO of 15 minutes**.

Q3. How is application scaling performed?

Application services primarily use horizontal scaling by increasing Kubernetes pod replicas.

Q4. When are storage resources reviewed?

Storage utilization is reviewed **weekly**.

Q5. Which Kubernetes security controls are implemented?

Role-Based Access Control (RBAC), Network Policies, Pod Security Standards, vulnerability scanning, and admission controller validation.

Q6. What activities are performed monthly?

Monthly activities include operating system patching, container image updates, monitoring review, security advisory review, and infrastructure cost analysis.

Q7. What should be verified after deployment?

Application availability, health endpoints, monitoring dashboards, log collection, smoke tests, and error rates.

Q8. How long are deployments monitored after release?

Deployments receive enhanced monitoring for the first **60 minutes** after completion.

Q9. Who responds to Priority 1 infrastructure incidents?

The on-call engineer, Incident Manager, and Platform Engineering response team.

Q10. What should trigger infrastructure expansion?

Sustained resource utilization, storage growth, increased API latency, or forecasted capacity exhaustion.

Revision History

Version	Date	Description
1.0	January 2024	Initial Infrastructure Guide
2.0	August 2025	Added Kubernetes and Monitoring Architecture
3.0	January 2026	Added Disaster Recovery, Scaling Strategy, Capacity Planning, and Operational Governance

Related Documents

- NovaDesk Developer Guide (TECH-DEV-001)
- NovaDesk REST API Documentation (TECH-API-001)
- Database Backup Manual (TECH-DB-001)
- Information Security Policy (COMP-SEC-003)
- Disaster Recovery Runbook (OPS-DR-001)
- Business Continuity Plan (OPS-BCP-002)